

RGB LED Effects

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1. RGB Colorful Light Overview

WS2812 (also known as NeoPixel) is a single-wire addressable RGB LED chip. Each LED has a built-in control IC, and only one data wire is needed to cascade and control hundreds of LEDs. This experiment drives the single onboard WS2812 (GPIO48), cycling through 7 rainbow colors, switching once per second while printing the current color name via serial.

Typical application scenarios:

Application Type	Example
Ambient lighting	Music-reactive lights, gaming ambient lights
Status indication	Color coding (red=fault, green=normal, blue=connecting)
Displays	LED matrix, digital tubes
Wearables	LED clothing, glowing accessories

2. Core Concepts

2.1 WS2812 Communication Protocol

The WS2812 uses a **single-wire return-to-zero code**: there is only one data wire, and different pulse widths represent 0 and 1.

0 code: 

0.4µs high + 0.85µs low

1 code: 

0.8µs high + 0.45µs low

RESET:  > 50µs

low > 50µs = latch signal

Each LED requires 24 bits of data (8 bits each for G-R-B); cascading N LEDs requires 24×N bits. The RESET signal latches the data in the shift register to the output.

2.2 WS2812 Color Format — GRB

The native color order of the WS2812 is **GRB** (Green-Red-Blue), which differs from the common RGB order:

#

G

R

B

```
RAINBOW = [  
  (150, 0, 0), # red:  G=0,  R=150, B=0  
  (0, 150, 0), # green: G=150, R=0,  B=0  
  (0, 0, 150), # blue:  G=0,  R=0,  B=150  
]
```

(150, 0, 0) in GRB format = G=150, R=0, B=0 → **dim green**. To write red, use (0, 150, 0): G=0, R=150, B=0.

2.3 Seven-Color Rainbow List

This experiment predefines 7 rainbow colors and cycles through them by array index:

Index	Color	GRB Value	Visual Effect
0	Red	(150, 0, 0)	Red
1	Orange	(150, 60, 0)	Orange (red+green)
2	Yellow	(150, 150, 0)	Yellow (red+green equal)
3	Green	(0, 150, 0)	Green
4	Cyan	(0, 150, 150)	Cyan (green+blue)
5	Blue	(0, 0, 150)	Blue
6	Purple	(120, 0, 150)	Purple (red+blue)

2.4 Core API — neopixel

```
from machine import Pin
from neopixel import NeoPixel

np = NeoPixel(Pin(48), 1)      # GPIO48, 1 LED
np[0] = (150, 0, 0)           # set LED 0 to red (GRB)
np.write()                    # send the color data to the LED
```

`np[0] = (g, r, b)` only modifies the buffer in memory; **you must call** `np.write()` **to actually send it to the LED.**

3. Hardware Dependencies

GPIO	Peripheral	Mode	Description
48	WS2812	<code>Pin.OUT</code> (NeoPixel)	Onboard single RGB LED, data input pin

The WS2812 on this board is onboard; no external components are required.

4. Project Architecture and Flow

```
Script execution (single-file .py)
|
|─ Config: LED_PIN=48, INTERVAL=1000ms
|─ RAINBOW = [7 GRB colors]           // predefined color array
|─ np = NeoPixel(Pin(48), 1)          // 1 WS2812
|─ color_index = 0                    // color index
|
|─ while True:                        // cycle switching
|   |─ print("Current color: ...")    // print the current color name
|   |─ np[0] = RAINBOW[color_index]  // set the color
|   |─ np.write()                    // send to the LED
|   |─ color_index = (index + 1) % 7 // cycle the index
|   └─ sleep_ms(1000)                // 1s interval
```

5. Source Code Details

Full source code: `5.Source Code` → `MicroPython` → `1.Basic Peripherals` → `8.RGB` → `8.RGB.py`

5.1 Parameter Configuration and Color Definition

```
from machine import Pin      # Pin class
from neopixel import NeoPixel # WS2812 driver
import time                  # delay
```

```
# Configuration parameters
LED_PIN = 48          # LED data pin
LED_NUM = 1           # number of LEDs
INTERVAL = 1000       # color switch interval (ms)

# Seven rainbow colors (GRB order, WS2812 native format)
RAINBOW = [
    (150, 0, 0),      # Red      (R=150, G=0, B=0)
    (150, 60, 0),     # Orange (R=150, G=60, B=0)
    (150, 150, 0),    # Yellow (R=150, G=150, B=0)
    (0, 150, 0),      # Green  (R=0, G=150, B=0)
    (0, 150, 150),    # Cyan   (R=0, G=150, B=150)
    (0, 0, 150),      # Blue   (R=0, G=0, B=150)
    (120, 0, 150),    # Purple (R=120, G=0, B=150)
]

COLOR_NAMES = ["Red", "Orange", "Yellow", "Green", "Cyan", "Blue", "Purple"]
```

5.2 Initialization and Cycle Switching

```
# Init NeoPixel (WS2812, GPIO48)
np = NeoPixel(Pin(LED_PIN), LED_NUM)

color_index = 0

while True:
    print("Current color: %s" % COLOR_NAMES[color_index]) # print color

    np[0] = RAINBOW[color_index] # write to buffer
    np.write()                  # send to LED

    color_index = (color_index + 1) % len(RAINBOW) # cycle index
    time.sleep_ms(INTERVAL)
```

`(color_index + 1) % len(RAINBOW)` wraps the index back to 0: 6→7→0→1... .

6. Operation Procedure

6.1 Flashing and Running

Thonny IDE steps: See the [1. Before Development](#) → [3. Thonny IDE](#) chapter.

6.2 Normal Phenomenon

- The RGB LED switches to one color per second:
red→orange→yellow→green→cyan→blue→purple→red, repeating

```
Current color: Red
Current color: Orange
Current color: Yellow
Current color: Green
Current color: Cyan
Current color: Blue
Current color: Purple
Current color: Red      ← cycle
...
```

```
Current color: Red
Current color: Orange
Current color: Yellow
Current color: Green
Current color: Cyan
Current color: Blue
Current color: Purple
Current color: Red
```

- The Shell prints the current color name every second

6.3 Verification Methods

Operation	Expected Result
Power up	LED starts with red, switches every second
Change <code>INTERVAL=1000</code> to <code>INTERVAL=200</code>	Switching speed increases 5 times
Comment out <code>np.write()</code>	LED color does not update
Modify a color value in the RAINBOW array	The corresponding color changes
Swap G and R in the RAINBOW array	Colors are clearly wrong (GRB vs RGB)

7. Common Problems

Symptom	Cause	Solution
<code>ImportError: no module named 'neopixel'</code>	The firmware does not include the <code>neopixel</code> module	Flash a MicroPython firmware that includes <code>neopixel</code>
LED does not light up	<code>np.write()</code> not called	Confirm <code>np.write()</code> is called after every color assignment
Color does not match the name (e.g., red becomes green)	Data order uses RGB instead of GRB	Swap the first and second values in the tuple
Uneven color brightness	The brightness values of the source colors are not calibrated	Adjust the channel values in the RAINBOW array
Serial prints fine but the LED stays white	Powered on before initialization; WS2812 defaults to random	Write black <code>(0,0,0)</code> once before starting the loop